Velocity–time graphs and vectors

LEARNING TARGET · R2 I can read a velocity–time graph and split a velocity arrow into two components.

A · KEY VOCABULARY

Velocity–time graph	The vertical axis shows *velocity*, not position. Read the label first.
Flat v–t line	Steady velocity — not necessarily stopped.
Area under the line	Height × width = displacement. Area below zero is negative displacement.
Components	The separate east and north changes that make up one diagonal displacement.

B · KEY EQUATIONS + UNITS

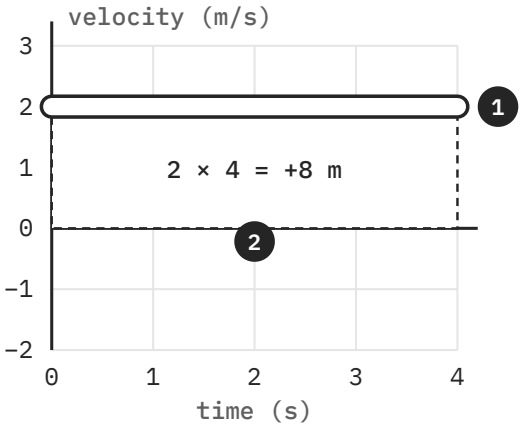
$$\Delta x = v \times t \quad \text{(area)}$$

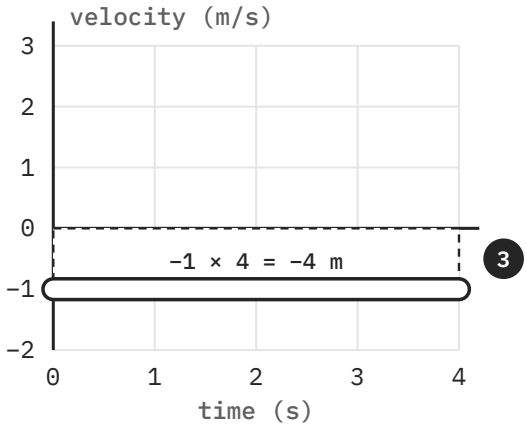
$$v_x = v \cos \theta$$
$$v_y = v \sin \theta$$

θ	measured from the +x axis
cos / sin	part next to / opposite θ

C · COLOR IT · AREA COUNTS THE CHANGE

Two velocity graphs. Right is +.





☐ **1 · PURPLE** Velocity lines

☐ **2 · BLUE** Area above zero: + displacement

☐ **3 · BLUE** Area below zero: stripes, − displacement

D · WORKED EXAMPLE

Velocity is +3 m/s for 2 s. Find the displacement and name its direction.

GIVEN	EQUATION	WORK	ANSWER
v = +3 m/s t = 2 s	Δx = v × t	Δx = (3 m/s)(2 s)	**+6 m** 6 m to the right

E · THE 2026-XJ THREAD

DAY 4 OF 22 · NEW TOOL: V–T GRAPHS + COMPONENTS

A velocity–time graph shows whether XJ's speed is holding steady. Splitting its velocity arrow into parts shows how much is aimed at Earth and how much carries it sideways.

Unit question: How fast is it moving, and where will it arrive?

UNIT 1 COLOR CODE ☐ green speeding up ☐ red slowing down ☐ yellow steady · balanced ☐ blue position · displacement ☐ purple velocity ☐ orange forces pencil · everything else